

iterate to get mean: $w_{k+1} = w_k - \frac{1}{k} (w_k - x_k)$

Stochastic Approximation (SA)

Stochastic + iterative

Prob: how to solve $g(w) = 0$ if the explicit formula of g is not given?

Algo (Robbins-Monro, RM)

let w^* to be a root of $g(w) = 0$.

then

Init: w_0

Iter: $w_{k+1} = w_k - a_k \tilde{g}(w_k, \eta_k)$ detectable

where

$$\tilde{g}(w_k, \eta_k) = g(w_k) + \eta_k$$

$$\sum a_k^2 < \infty, \sum a_k = \infty. \quad (\text{e.g. } a_k = \frac{1}{k})$$

$$\Rightarrow w_{k+1} \xrightarrow{P} w^*$$

Stochastic Gradient Descent (SGD)

$$\textcircled{1} \text{ GD: } w_{k+1} = w_k - \alpha_k \mathbb{E}[\nabla f(w_k, x)]$$

$$\textcircled{2} \text{ BGD: } w_{k+1} = w_k - \alpha_k \sum_{i=1}^n \nabla_{w_k} f(w_k, x_i) \quad \text{all}$$

$$\textcircled{3} \text{ MBGD: } w_{k+1} = w_k - \alpha_k \frac{1}{m} \sum_{j \in I_k} \nabla_{w_k} f(w_k, x_j) \quad \text{some}$$

$$\textcircled{4} \text{ SGD: } w_{k+1} = w_k - \alpha_k \nabla_{w_k} f(w_k, x_k) \quad \text{one}$$